

Related Work Research Package for the TreeMAPPO Draft

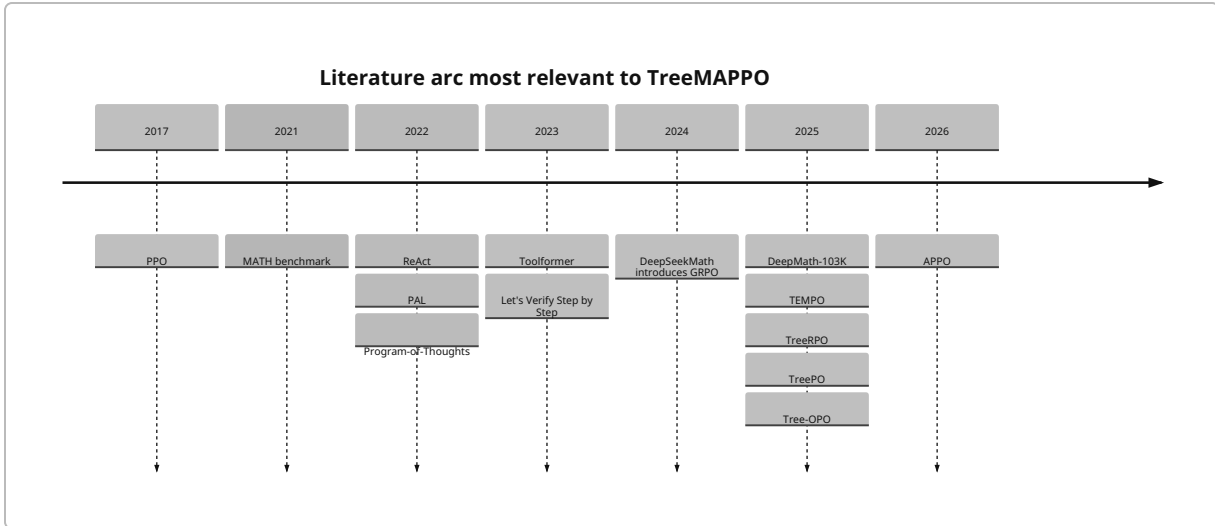
Executive summary

The draft positions **TreeMAPPO** as a reference-free, on-policy method for fine-grained credit assignment in mathematical reasoning agents, with four core claims: token-level trajectory branching, posterior inference over signed segment contribution and uncertainty, dense token-level advantages, and applicability to mixed text-plus-Python-tool trajectories. The most relevant prior work splits into five strands: trajectory-level RL/RLVR such as PPO and GRPO-style training; process-supervision and verifier-based methods; tree- or prefix-structured credit assignment methods such as TEMPO, TreeRPO, TreePO, and Tree-OPO; tool-augmented reasoning methods such as PAL, PoT, ReAct, and Toolformer; and benchmark/data resources such as MATH, NuminaMath, DeepMath-103K, AMC23, GaoKaoMath, and CollegeMath. Relative to this landscape, the draft's clearest novelty is not "tree structure" alone, but the combination of **active token-level branching** with an **explicit uncertainty-aware posterior credit model** for agentic reasoning trajectories, which distinguishes it from natural shared-prefix methods, fixed-segment tree methods, and off-policy teacher-tree approaches. Because the current draft still contains placeholder result tables, the strongest related-work framing should emphasize methodological differentiation, benchmark coverage, and limitations rather than premature superiority claims. [filecite turn0file0](#) 1

What the draft is already claiming

From the uploaded draft, the paper's central problem is **sparse end-of-trajectory supervision** for long mathematical reasoning trajectories that may include natural-language reasoning, fenced Python tool calls, tool responses, and terminal boxed answers. The proposed method, TreeMAPPO, builds trees from sampled rollouts, branches at selected token positions, infers segment-level contribution and uncertainty with a MAP posterior model, and converts those estimates into token-level advantages for LoRA fine-tuning. The intended comparisons are against a base model, a GRPO-style trajectory-level baseline, a TEMPO-style spontaneous branching ablation, and a TreeRPO-style heuristic credit ablation; the experimental plan covers both tool and no-tool settings across DeepMath, MATH, NuminaMath, AMC2023, GaoKao Math 2024, and CollegeMath. The draft's current results section is still placeholder text, so the related-work section should explicitly frame novelty by **what kind of credit signal is inferred, where branching occurs, and whether the method depends on teacher trees, process labels, or heuristic estimates** rather than by claimed empirical dominance. [filecite turn0file0](#)

The most important topics to foreground in the literature are therefore: RL with verifiable rewards for reasoning; process supervision versus outcome-only supervision; tree-structured or prefix-aware credit assignment; off-policy versus on-policy tree construction; uncertainty-aware branching; and tool-augmented mathematical reasoning. Those themes track the draft's actual contribution much better than a generic "reasoning with LLMs" survey would. [filecite turn0file0](#) 2



The timeline above captures the progression from general RL optimization and math benchmarks to tool use, process verification, RLVR, and then explicitly tree-structured or agentic credit-assignment methods. That progression supports a related-work narrative in which TreeMAPPO is best presented as the next step after RLVR and tree-structured rollout methods, rather than as a general-purpose reasoning paper. ³

Paper-ready Related Work section

Reinforcement learning for mathematical reasoning is increasingly organized around **verifiable outcome rewards**, where correctness can be checked automatically from a final answer. In this setting, PPO remains the canonical policy-optimization baseline because it yields token-level advantages through a critic \cite{schulman2017ppo}; however, critic training adds complexity and may be unstable for long-form reasoning. More recent reasoning systems therefore often adopt critic-free or reduced-critic variants such as GRPO, introduced in DeepSeekMath, which use group-relative outcome signals while preserving the practical appeal of verifiable rewards \cite{shao2024deepseekmath}. The central weakness of this family is the same weakness emphasized by our setting: when reward is attached primarily at the trajectory level, long reasoning chains receive sparse and weakly localized feedback, so the learner may reinforce or suppress many irrelevant tokens together with the truly decisive ones. ⁴

A parallel line of work argues that better reasoning requires **denser supervision over intermediate steps** rather than only final correctness. The clearest representative is *Let's Verify Step by Step* \cite{lightman2023letsverify}, which showed that process supervision can outperform pure outcome supervision on MATH by training verifiers from step-level human labels. Subsequent work on process reward models further explores how to score intermediate reasoning and use those signals for selection or optimization \cite{zhang2025lessonsprm,khalifa2025thinkprm}. These papers are highly relevant because they establish the value of fine-grained credit; however, they rely on either annotated step labels or learned verifiers, whereas TreeMAPPO is intended to remain **reference-free** and to infer local credit directly from the structure of sampled rollouts. That distinction matters both practically, because step-level labels are expensive, and methodologically, because the draft's contribution is not better verification per se but better **credit assignment under outcome-only supervision**. ⁵

The closest literature is the recent wave of **tree-structured or prefix-aware RLVR methods**. TEMPO converts a group of sampled completions into a prefix tree and augments GRPO with branch-gated temporal-difference corrections \cite{tran2025tempo}. Its key insight is that shared prefixes contain nonparametric estimates of local value, allowing token-specific corrections without a learned critic. Yet TEMPO depends on **naturally occurring prefix overlap** among independent samples; it does not actively create new continuations at chosen decision points. TreeRPO takes a more explicit tree-sampling route, estimating step-level reward expectations from tree rollouts and forming step-wise comparison groups without a separate reward model \cite{yang2025treerpo}. TreePO likewise treats decoding as a tree search, but emphasizes systems efficiency through shared-prefix reuse, fixed-length segment decoding, dynamic divergence, and tree-based segment-level advantages \cite{li2025treepo}. Tree-OPO extends the family in another direction by using **teacher-generated MCTS prefixes** and staged, prefix-conditioned advantage estimation in an off-policy setting \cite{huang2025treeopo}. Across these works, the main design axes are where branches come from, what unit receives credit, and whether advantages are estimated heuristically, empirically, or with a learned/model-based procedure. 6

Against that background, TreeMAPPO’s most credible point of differentiation is not simply “using a tree,” but the specific combination of **active token-level branching** and **uncertainty-aware posterior credit inference**. Compared with TEMPO, the method can be framed as replacing incidental shared-prefix overlap with guided branching at selected token positions. Compared with TreeRPO and TreePO, it can be framed as relaxing fixed step or segment boundaries and replacing win-rate or subgroup-style credit heuristics with a posterior over signed segment contribution and uncertainty. Compared with Tree-OPO, it can be framed as remaining on-policy and avoiding offline teacher trees. This framing is especially strong if the paper avoids overstating the scope of the Bayesian language: the important claim is that uncertainty is modeled explicitly and then reused both for branching and for advantage construction, which is a sharper methodological distinction than simply saying the method is “statistically grounded.” \filecite{turn0file0} 6

A second neighboring literature concerns **tool-augmented reasoning**, which is especially relevant because the draft’s trajectories may interleave plain-text thoughts, fenced Python code, tool responses, and final boxed answers. PAL and Program-of-Thoughts argued early that language models can delegate arithmetic or symbolic computation to Python, substantially improving numerical reasoning by disentangling decomposition from execution \cite{gao2022pal,chen2022pot}. ReAct generalized this idea to interleaved reasoning-and-acting trajectories, while Toolformer showed that models can learn when and how to call tools with limited demonstrations \cite{yao2022react,schick2023toolformer}. These papers motivate the draft’s agent protocol, but they mainly address **inference-time capability and tool invocation**, not training-time credit assignment. The closer point of contact is APPO, which studies branch placement and procedural credit in agentic RL, arguing that influential decisions are spread throughout a trajectory rather than concentrated at tool-call boundaries \cite{wang2026appo}. This makes APPO the most natural recent comparator for the tool-enabled setting, although the draft can still distinguish itself by emphasizing posterior segment inference instead of procedure-level advantage scaling. 7

The benchmark landscape also matters for how the related-work section should be written. MATH remains the canonical competition-math benchmark for stepwise reasoning \cite{hendrycks2021math}, while more recent resources such as NuminaMath and DeepMath-103K were explicitly curated to support large-scale post-training or RL with verifiable answers \cite{numina2024dataset,he2025deepmath103k}. The draft’s training design—using DeepMath, MATH, and NuminaMath for in-distribution training/validation and holding out AMC2023, GaoKao Math 2024, and CollegeMath for broader testing—fits current practice in aiming to separate training corpora from harder or structurally different evaluation sets. At the same time,

that design introduces an opportunity for the related-work section: it can argue that prior tree-credit papers are often evaluated on standard text-only reasoning suites, whereas TreeMAPPO is intended for both **tool and no-tool** mathematical reasoning. That broader framing is important because many existing gains may be benchmark-specific, and the agentic setting changes both the action space and the attribution problem. turn0file0 8

Overall, the literature suggests a concise positioning statement: trajectory-level RLVR provides scalable outcome supervision but weak credit localization; process supervision offers dense guidance but requires additional labels or verifiers; recent tree methods show that structural rollouts can densify credit, yet they vary in whether branching is incidental, fixed-granularity, heuristic, off-policy, or agent-specific. TreeMAPPO should therefore be presented as an attempt to unify the strongest parts of these threads for mathematical agents: **reference-free RLVR, explicit tree exploration, token-level branch placement, uncertainty-aware local credit, and compatibility with tool-interleaved trajectories**. That is the gap the current draft most plausibly addresses. turn0file0 9

Comparison of key papers

The table below compares the papers most useful for positioning TreeMAPPO. Headline performance figures are the authors’ own reported results and should **not** be read as directly comparable across different models, benchmarks, or evaluation protocols.  10

Year	Paper	Method	Main dataset or setting	Headline performance reported by authors	Relevance to TreeMAPPO
2017	PPO	Critic-based policy optimization with clipped objective	Atari, robotics control	Foundational RL algorithm; simpler than TRPO and strong sample efficiency in original benchmarks	Base conceptual ancestor for policy optimization and token-level advantage ideas.  11
2021	MATH	Benchmark paper	12,500 competition math problems	Introduced a hard benchmark and argued scaling alone was insufficient	Canonical math-reasoning benchmark that motivates fine-grained credit.  12

Year	Paper	Method	Main dataset or setting	Headline performance reported by authors	Relevance to TreeMAPPO
2022	PAL	Program-aided tool use with Python execution	13 reasoning/algorithmic tasks	Surpassed PaLM-540B CoT on GSM8K by 15 absolute points in the reported setup	Strong antecedent for the draft's Python-tool trajectory format. ¹³
2022	Program-of-Thoughts	Programmatic reasoning with external execution	8 numerical/financial reasoning tasks	About 12% average gain over CoT across evaluated tasks	Supports the claim that tool-executed computation materially changes reasoning trajectories. ¹⁴
2022	ReAct	Interleaved reasoning and acting	QA plus interactive environments	Improved success in ALFWorld by 34% and WebShop by 10% over baselines reported in the paper	Important precedent for mixed reasoning-action trajectories, though not a training-time credit method. ¹⁵
2023	Toolformer	Self-supervised tool-use learning	API/tool-augmented LM tasks	Consistent zero-shot improvements across downstream tasks	Establishes that tool calls can be learned as part of sequence modeling, strengthening the agentic framing. ¹⁶

Year	Paper	Method	Main dataset or setting	Headline performance reported by authors	Relevance to TreeMAPPO
2023	Let's Verify Step by Step	Process supervision with step-level labels	MATH subset, PRM800K	Process-supervised model solved 78% of a representative MATH subset	Key adjacent line showing why dense intermediate supervision matters, even though it is not reference-free. ¹⁷
2024	DeepSeekMath	GRPO-style RLVR for math reasoning	MATH and math training corpus	51.7% on MATH; 60.9% with self-consistency over 64 samples	Canonical open reference for GRPO-style trajectory-level RLVR baseline. ¹⁸
2025	DeepMath-103K	RL-oriented dataset/resource	103K verifiable, decontaminated math problems	Authors report significant improvements on challenging math benchmarks after training	Highly relevant resource because the draft trains on DeepMath and emphasizes RL-suitable verifiable data. ¹⁹
2025	TEMPO	Prefix-tree credit from naturally shared prefixes	MATH, MedQA, GSM-HARD, AMC23, MedMCQA, MMLU-Medical	Outperformed PPO and GRPO with roughly similar wall-clock time in reported experiments	Closest baseline for shared-prefix, critic-free token-local credit; differs because it does not actively branch. ²⁰

Year	Paper	Method	Main dataset or setting	Headline performance reported by authors	Relevance to TreeMAPPO
2025	TreeRPO	Explicit tree sampling with step-level reward expectations	Qwen-2.5-Math test benchmarks	Reported average Pass@1 increase from 19.0% to 35.5%; 2.9 points above GRPO	Direct comparator on tree-structured dense rewards; TreeMAPPO differs in token-level, uncertainty-aware posterior credit. ²¹
2025	TreePO	Dynamic tree sampling with fixed-length segments and tree-aware advantages	Reasoning benchmarks	Reported 22–43% GPU-hour savings and up to 40% trajectory-level compute reduction	Important efficiency-oriented tree baseline; useful contrast because TreeMAPPO prioritizes probabilistic credit estimation over systems efficiency. ²²
2025	Tree-OPO	Off-policy MCTS-prefix training with staged advantages	Multistep reasoning	Reported stabilization benefits and improved compositional reward structure; early-stage empirical results	Good contrast case for prefix-aware off-policy training from teacher trees, unlike TreeMAPPO’s on-policy construction. ²³

Year	Paper	Method	Main dataset or setting	Headline performance reported by authors	Relevance to TreeMAPPO
2026	APPO	Fine-grained branch placement and procedure-level scaling for agentic RL	13 agentic benchmarks	Nearly 4-point average improvement over strong agentic RL baselines	Most relevant recent agentic comparator; especially important for the tool-enabled setting. ²⁴

Recommended citations and BibTeX

The following set is the most defensible 12-paper citation pack for the current draft. It balances foundational RL, benchmark/data papers, process supervision, tree-structured credit assignment, and tool-augmented reasoning. The clickable citations point to primary sources. ²⁵

1. [schulman2017ppo](#) — foundational optimizer for policy-gradient RL. ¹¹
2. [hendrycks2021math](#) — canonical competition-math benchmark. ¹²
3. [gao2022pal](#) — Python-assisted reasoning for mathematical tasks. ¹³
4. [chen2022pot](#) — reasoning-as-programs with external execution. ¹⁴
5. [yao2022react](#) — interleaved reasoning and acting. ¹⁵
6. [schick2023toolformer](#) — self-supervised tool-use learning. ¹⁶
7. [lightman2023letsverify](#) — process supervision and PRM800K. ¹⁷
8. [shao2024deepseekmath](#) — DeepSeekMath and GRPO. ¹⁸
9. [he2025deepmath103k](#) — RL-oriented math dataset used in the draft. ¹⁹
10. [tran2025tempo](#) — shared-prefix tree credit assignment. ²⁰
11. [yang2025treerpo](#) — explicit tree-sampled step-level rewards. ²¹
12. [li2025treepo](#) — tree-structured rollout efficiency and segment-level advantages. ²²
13. [huang2025treeopo](#) — off-policy tree-guided advantage estimation. ²³
14. [wang2026appo](#) — fine-grained agentic branching and procedural credit. ²⁴

```
@article{schulman2017ppo,
  title   = {Proximal Policy Optimization Algorithms},
  author  = {Schulman, John and Wolski, Filip and Dhariwal, Prafulla and
Radford, Alec and Klimov, Oleg},
  journal = {arXiv preprint arXiv:1707.06347},
  year    = {2017},
  url     = {https://arxiv.org/abs/1707.06347}
}

@article{hendrycks2021math,
  title   = {Measuring Mathematical Problem Solving With the MATH Dataset},
```



```

    author = {Hendrycks, Dan and Burns, Collin and Kadavath, Saurav and Arora,
Akul and Basart, Steven and Tang, Eric and Song, Dawn and Steinhardt, Jacob},
    journal = {Advances in Neural Information Processing Systems},
    year = {2021},
    url = {https://arxiv.org/abs/2103.03874}
}

@article{gao2022pal,
    title = {PAL: Program-aided Language Models},
    author = {Gao, Luyu and Madaan, Aman and Zhou, Shuyan and Alon, Uri and Liu,
Pengfei and Yang, Yiming and Callan, Jamie and Neubig, Graham},
    journal = {arXiv preprint arXiv:2211.10435},
    year = {2022},
    url = {https://arxiv.org/abs/2211.10435}
}

@article{chen2022pot,
    title = {Program of Thoughts Prompting: Disentangling Computation from
Reasoning for Numerical Reasoning Tasks},
    author = {Chen, Wenhui and Ma, Xueguang and Wang, Xinyi and Cohen, William
W.},
    journal = {arXiv preprint arXiv:2211.12588},
    year = {2022},
    url = {https://arxiv.org/abs/2211.12588}
}

@article{yao2022react,
    title = {ReAct: Synergizing Reasoning and Acting in Language Models},
    author =
{Yao, Shunyu and Zhao, Jeffrey and Yu, Dian and Du, Nan and Shafran, Izhak and
Narasimhan, Karthik and Cao, Yuan},
    journal = {arXiv preprint arXiv:2210.03629},
    year = {2022},
    url = {https://arxiv.org/abs/2210.03629}
}

@article{schick2023toolformer,
    title = {Toolformer: Language Models Can Teach Themselves to Use Tools},
    author = {Schick, Timo and Dwivedi-Yu, Jane and Dess{\`i}, Roberto and
Raileanu, Roberta and Lomeli, Maria and Zettlemoyer, Luke and Cancedda, Nicola
and Scialom, Thomas},
    journal = {arXiv preprint arXiv:2302.04761},
    year = {2023},
    url = {https://arxiv.org/abs/2302.04761}
}

@article{lightman2023letsverify,
    title = {Let's Verify Step by Step},

```

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    author = {Lightman, Hunter and Kosaraju, Vineet and Burda, Yura and Edwards,
Harri and Baker, Bowen and Lee, Teddy and Leike, Jan and Schulman, John and
Sutskever, Ilya and Cobbe, Karl},
    journal = {arXiv preprint arXiv:2305.20050},
    year = {2023},
    url = {https://arxiv.org/abs/2305.20050}
}

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```

@article{shao2024deepseekmath,
    title = {DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open
Language Models},
    author = {Shao, Zhihong and Wang, Peiyi and Zhu, Qihao and Xu, Runxin and
Song, Junxiao and Bi, Xiao and Zhang, Haowei and Zhang, Mingchuan and Li, Y. K.
and Wu, Y. and Guo, Daya},
    journal = {arXiv preprint arXiv:2402.03300},
    year = {2024},
    url = {https://arxiv.org/abs/2402.03300}
}

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@article{he2025deepmath103k,
    title = {DeepMath-103K: A Large-Scale, Challenging, Decontaminated, and
Verifiable Mathematical Dataset for Advancing Reasoning},
    author =
{He, Zhiwei and Liang, Tian and Xu, Jiahao and Liu, Qiuzhi and Chen, Xingyu and
Wang, Yue and Song, Linfeng and Yu, Dian and Liang, Zhenwen and Wang, Wenxuan
and Zhang, Zhuosheng and Wang, Rui and Tu, Zhaopeng and Mi, Haitao and Yu,
Dong},
    journal = {arXiv preprint arXiv:2504.11456},
    year = {2025},
    url = {https://arxiv.org/abs/2504.11456}
}

```

```

@article{tran2025tempo,
    title = {Exploiting Tree Structure for Credit Assignment in RL Training of
LLMs},
    author = {Tran, Hieu and Yao, Zonghai and Yu, Hong},
    journal = {arXiv preprint arXiv:2509.18314},
    year = {2025},
    url = {https://arxiv.org/abs/2509.18314}
}

```

```

@article{yang2025treerpo,
    title = {TreeRPO: Tree Relative Policy Optimization},
    author = {Yang, Zhicheng and Guo, Zhijiang and Huang, Yinya and Liang,
Xiaodan and Wang, Yiwei and Tang, Jing},
    journal = {arXiv preprint arXiv:2506.05183},
    year = {2025},
    url = {https://arxiv.org/abs/2506.05183}
}

```

```

}

@article{li2025treepo,
  title = {TreePO: Bridging the Gap of Policy Optimization and Efficacy and Inference Efficiency with Heuristic Tree-based Modeling},
  author = {Li, Yizhi and Gu, Qingshui and Wen, Zhoufutu and Li, Ziniu and Xing, Tianshun and Guo, Shuyue and Zheng, Tianyu and Zhou, Xin and Qu, Xingwei and Zhou, Wangchunshu and Zhang, Zheng and Shen, Wei and Liu, Qian and Lin, Chenghua and Yang, Jian and Zhang, Ge and Huang, Wenhao},
  journal = {arXiv preprint arXiv:2508.17445},
  year = {2025},
  url = {https://arxiv.org/abs/2508.17445}
}

@article{huang2025treepo,
  title = {Tree-OP0: Off-policy Monte Carlo Tree-Guided Advantage Optimization for Multistep Reasoning},
  author = {Huang, Bingning and Nguyen, Tu and Zimmer, Matthieu},
  journal = {arXiv preprint arXiv:2509.09284},
  year = {2025},
  url = {https://arxiv.org/abs/2509.09284}
}

@article{wang2026appo,
  title = {APPO: Agentic Procedural Policy Optimization},
  author = {Wang, Xucong and Ma, Ziyu and Wang, Yong and Ji, Yuxiang and Yang, Shidong and Chen, Guanhua and Wang, Pengkun and Chu, Xiangxiang},
  journal = {arXiv preprint arXiv:2606.12384},
  year = {2026},
  url = {https://arxiv.org/abs/2606.12384}
}

```

A final drafting note: because the uploaded manuscript already contains a short Related Work section and placeholder result tables, the cleanest revision strategy is to replace that section with the paper-ready text above, keep claims tightly calibrated, and avoid any sentence that implies empirical superiority until the draft's own tables are populated with measured numbers. fileciteturn0file0

1 3 4 10 11 25 Proximal Policy Optimization Algorithms

https://arxiv.org/abs/1707.06347?utm_source=chatgpt.com

2 5 17 Let's Verify Step by Step

https://arxiv.org/abs/2305.20050?utm_source=chatgpt.com

6 20 Exploiting Tree Structure for Credit Assignment in RL Training of LLMs

https://arxiv.org/abs/2509.18314?utm_source=chatgpt.com

7 13 PAL: Program-aided Language Models

https://arxiv.org/abs/2211.10435?utm_source=chatgpt.com

8 12 Measuring Mathematical Problem Solving With the MATH Dataset

https://arxiv.org/abs/2103.03874?utm_source=chatgpt.com

9 18 DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

https://arxiv.org/abs/2402.03300?utm_source=chatgpt.com

14 Program of Thoughts Prompting: Disentangling Computation from Reasoning for Numerical Reasoning Tasks

https://arxiv.org/abs/2211.12588?utm_source=chatgpt.com

15 ReAct: Synergizing Reasoning and Acting in Language Models

https://arxiv.org/abs/2210.03629?utm_source=chatgpt.com

16 Toolformer: Language Models Can Teach Themselves to Use Tools

https://arxiv.org/abs/2302.04761?utm_source=chatgpt.com

19 DeepMath-103K: A Large-Scale, Challenging, Decontaminated, and Verifiable Mathematical Dataset for Advancing Reasoning

https://arxiv.org/abs/2504.11456?utm_source=chatgpt.com

21 TreeRPO: Tree Relative Policy Optimization

https://arxiv.org/abs/2506.05183?utm_source=chatgpt.com

22 TreePO: Bridging the Gap of Policy Optimization and Efficacy and Inference Efficiency with Heuristic Tree-based Modeling

https://arxiv.org/abs/2508.17445?utm_source=chatgpt.com

23 Tree-OPO: Off-policy Monte Carlo Tree-Guided Advantage Optimization for Multistep Reasoning

https://arxiv.org/abs/2509.09284?utm_source=chatgpt.com

24 APPO: Agentic Procedural Policy Optimization

https://arxiv.org/abs/2606.12384?utm_source=chatgpt.com